

JEPA-Q2D2: A Low-Bitrate Speech Codec with Emergent Cross-Lingual Structure

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Abstract—We present a 1.6 kbps neural speech codec that pairs a JEPA (Joint-Embedding Predictive Architecture) encoder with Q2D2, a geometry-aware two-dimensional lattice quantizer, and is trained without adversarial losses. Q2D2 quantizes latent dimension pairs on a rhombic lattice with per-pair learned affine transforms; a frozen WavLM perceptual loss is used during decoder training. Under an identical internal ablation protocol, Q2D2 improves over finite scalar quantization (FSQ) by +0.37 PESQ at a matched bitrate. On a separate fixed 50-utterance LibriLight evaluation, the codec reaches PESQ 2.53 and ESTOI 0.80 at 1.6 kbps, exceeding EnCodec at a comparable operating point but trailing Mimi on ESTOI. We further show that SIGReg, a Gaussianizing regularizer applied to the Stage-1 encoder, prevents Q2D2 collapse in a more aggressive 25 Hz / cd32 regime. Despite English-only LibriLight training, the codec’s pre-quantization encoder features on held-out FLEURS speech (five unseen non-English languages plus a disjoint English subset) reach 85.4% 5-NN language separability, higher than every neural-codec baseline (Mimi, EnCodec, DAC) and every multilingual SSL model we test (XLS-R-300m, mHuBERT-147), approaching uncompressed English wav2vec2-base.

Index Terms—neural speech codec, vector quantization, lattice quantization, self-supervised learning, cross-lingual representation

I. INTRODUCTION

Discrete speech codecs have become the interface between waveforms and language models: an autoregressive (AR) text-to-speech (TTS) system predicts codec tokens, which a decoder renders to audio. This use case imposes two demands that differ from classical speech coding. First, the token stream must be *short* (an AR model pays a quadratic cost in sequence length), so the codec should operate at a low frame rate and a low bitrate. Second, training must be *stable and simple*, because codec quality directly bounds downstream TTS quality and because practitioners increasingly train many codec variants.

Most recent neural codecs [1]–[4] answer the first demand with residual vector quantization (RVQ) and the second with multi-scale GAN discriminators. Both choices have costs. RVQ and learned vector quantization (VQ) are prone to codebook collapse and require commitment losses and careful initialization. GAN discriminators introduce a non-stationary adversarial objective whose instability is well documented. Finite scalar quantization (FSQ) [5] removes codebook learning but

quantizes each latent dimension independently on an axis-aligned grid, ignoring correlations between dimensions.

We study a codec built from two components chosen to address these costs directly. The encoder is a JEPA model [6], a self-supervised architecture that predicts masked latent features rather than reconstructing inputs; it is trained from scratch on English-only LibriLight. The quantizer is Q2D2 [7], which groups latent dimensions into pairs and snaps each pair to a rhombic (A_2) lattice after a per-pair learned affine transform. Because lattice quantization is provably efficient only for sources that match the lattice geometry, we additionally shape the encoder distribution with SIGReg, a Cramér–Wold Gaussianizing regularizer, and we train the decoder with a frozen-WavLM perceptual loss and no adversarial losses.

The main contributions of this paper are:

- 1) **A JEPA encoder paired with Q2D2, and the SIGReg co-design the pairing needs at aggressive operating points.** Q2D2 was introduced as an audio codec with a conventional convolutional encoder. We pair it with a JEPA encoder and find that at an aggressive 25 Hz, 32-dimensional operating point the codec collapses (ESTOI -0.004) unless the encoder distribution is Gaussianized with SIGReg, which restores it to ESTOI 0.79 (Fig. 2); at the gentler 12.5 Hz, 64-dimensional setting of our main model the encoder is already well-conditioned and SIGReg is not required. We train the codec with no GAN discriminator and report affine and codebook-usage diagnostics confirming the quantizer trains healthily (Section III-C: per-pair affine condition number 1.18, determinants 0.99 ± 0.002 , packed-vocabulary use 17.0%). An in-pipeline comparison reproduces a Q2D2-over-FSQ advantage on JEPA features (+0.37 PESQ at matched 1.6 kbps).
- 2) **Cross-lingual acoustic structure emerges in the codec encoder.** On held-out FLEURS speech (five unseen non-English languages plus a disjoint English subset), pre-quantization encoder features organize the six languages into frame-level clusters whose separability exceeds every neural-codec baseline and every multilingually-trained SSL model we test, while the codec itself remains a 1.6 kbps reconstruction system.

We are explicit about scope. We do not claim state-of-the-

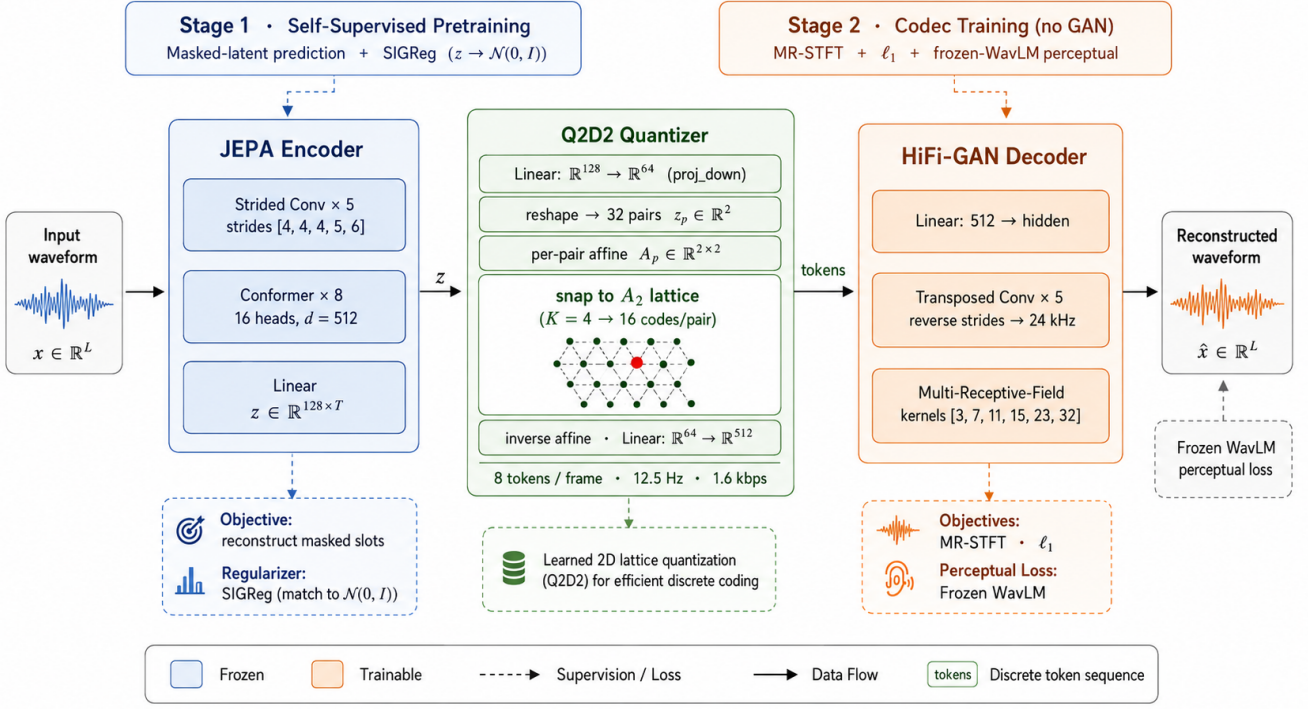


Fig. 1. Codec architecture. A JEPA encoder maps the waveform to a 128-dimensional latent z ; Q2D2 quantizes dimension pairs on a rhombic lattice with per-pair learned affine transforms; a HiFi-GAN decoder reconstructs audio. SIGReg shapes the encoder distribution during Stage 1; Stage 2 uses multi-resolution STFT, ℓ_1 and frozen-WavLM perceptual losses, with no GAN discriminator.

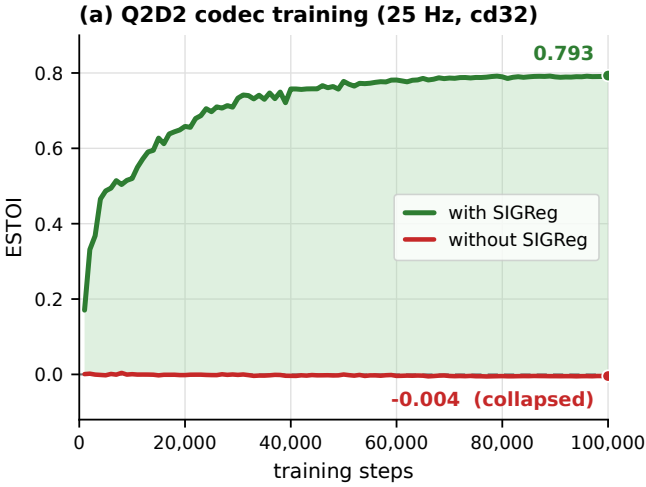


Fig. 2. SIGReg distribution-quantizer co-design. Two Q2D2 codecs at an aggressive 25 Hz / 32-dimensional operating point, identical except for the SIGReg weight used to train the Stage 1 encoder. Without SIGReg ($\lambda_{\text{sigreg}}=0$, red) the codec never leaves the floor; with SIGReg ($\lambda_{\text{sigreg}}=0.05$, green) the same codec trains normally to ESTOI 0.79. See Section III-D.

art reconstruction: on a matched protocol, Mimi attains higher ESTOI at a lower bitrate (Section III-B). Our contributions are a quantizer recipe and an analysis of what the encoder represents, the kind of “novel insight from evaluating existing approaches” that we believe is useful to the community.

II. METHOD

A. Architecture

Fig. 1 gives an overview of the codec.

Encoder. The encoder is a JEPA convolutional-Conformer network with channel widths [64, 128, 256, 384, 512, 512], eight residual blocks per stage and eight Conformer blocks (16 heads), producing a 128-dimensional latent. Two stride configurations are used: [4, 4, 4, 5, 6] (hop 1920, 12.5 Hz frames, 80 ms/frame) for our main model, and [4, 4, 4, 5, 3] (hop 960, 25 Hz) for a higher-frame-rate variant. The encoder is pre-trained in Stage 1 with a predictive objective on masked latent spans and an exponential-moving-average target encoder.

Quantizer. Q2D2 [7] reshapes the D -dimensional latent into $D/2$ coordinate pairs. Each pair passes through a learned 2×2 affine transform (identity-initialized), is snapped to the nearest point of a fixed rhombic (A_2) grid with $K=4$ levels per axis (16 codes per pair), and is mapped back through the inverse affine; gradients use a straight-through estimator. Our main model uses code dimension 64 (abbreviated *cd64* in tables and figures; *cd32* denotes the matched-bitrate 32-dimensional variant): 32 pairs are packed in groups of four into 8 tokens per frame, each with a vocabulary of $16^4 = 65,536$. At 12.5 Hz this is 100 tokens/s and $8 \cdot 16 \cdot 12.5 = 1.6$ kbps.

Decoder. A HiFi-GAN-style decoder with multi-receptive-field blocks (kernels [3, 7, 11, 15, 23, 32]) maps quantized latents back to 24 kHz audio. A learned 1×1 projection adapts the code dimension to the decoder width.

B. Distribution-quantizer co-design with SIGReg

Lattice quantizers are efficient for sources whose density matches the lattice geometry; for the A_2 grid, an isotropic source is the favorable case. We can therefore shape the encoder distribution with SIGReg, a Cramér–Wold / sliced-Wasserstein regularizer that penalizes deviation of the encoder feature distribution from $\mathcal{N}(0, I)$, applied during Stage 1 encoder training. Whether SIGReg is required depends on the operating point. At an aggressive 25 Hz, 32-dimensional setting a Q2D2 codec built on a plain EMA-target encoder collapses (ESTOI -0.004), while the same codec built on a SIGReg-shaped encoder trains normally (ESTOI 0.79; Section III-D). At the gentler 12.5 Hz, 64-dimensional setting of our main model the encoder is sufficiently well-conditioned that SIGReg is not needed.

C. Discriminator-free training

Stage 2 trains the decoder (and fine-tunes the encoder at $0.1\times$ the decoder learning rate via a gradient-scaling hook) with a multi-resolution STFT loss ($\lambda_{\text{stft}}=2.0$), an ℓ_1 waveform loss, and an ℓ_1 perceptual loss on layers 6 and 12 of a frozen WavLM-Base ($\lambda_{\text{perc}}=0.1$). No GAN discriminator is used; the discriminator graph is never constructed. We also use a two-level NaN guard that, in addition to the usual forward-loss check, inspects parameter gradients after the backward pass and skips the optimizer step if any are non-finite. This safeguard emerged from debugging a lattice-quantizer variant whose straight-through gradients could silently corrupt all parameters through gradient-norm clipping.

D. Training data and protocol

Stage 1 and Stage 2 use English-only LibriLight (medium subset, $\approx 5.8\text{k}$ hours). Optimization uses AdamW ($\beta = (0.8, 0.99)$), weight decay 10^{-3} , a cosine learning-rate schedule with linear warmup, batch size 4, and bfloat16 on a single A100-80GB GPU. The main model trains for 400k Stage 2 steps.

III. EXPERIMENTS

A. Evaluation protocol

All cross-system comparisons use a single fixed protocol: 50 LibriLight utterances (seed 42, $\leq 10\text{s}$ each), wideband PESQ on 16 kHz signals, and ESTOI (extended STOI). Speaker similarity is the cosine similarity of WavLM-base-plus-sv x-vectors. EnCodec, DAC and Mimi baselines are run through the identical pipeline on the same utterances. A separate internal evaluation logged during training (on a smaller, easier held-out set) is used *only* for the FSQ-vs-Q2D2 ablation, where both arms share that identical internal protocol; the two protocols are never mixed.

B. Reconstruction quality

Table I reports the main comparison. Among low-bitrate systems, our 12.5 Hz codec attains PESQ 2.53 and ESTOI 0.80 at 1.6 kbps, clearly exceeding EnCodec at 1.5 kbps on both metrics (+0.91 PESQ, +0.15 ESTOI). Mimi, at a lower

TABLE I
RECONSTRUCTION QUALITY, FIXED 50-UTTERANCE PROTOCOL. ESTOI IS EXTENDED STOI; ALL SYSTEMS USE THE IDENTICAL PIPELINE. TOKEN RATES FOR RVQ BASELINES COUNT ALL CODEBOOKS.

System	kbps	tok/s	PESQ	ESTOI
DAC	8.0	~ 675	4.46	0.985
EnCodec	6.0	~ 600	2.84	0.819
Ours (cd128, 12.5 Hz, teacher)	2.9	237.5	3.14	0.874
Ours (cd64, 12.5 Hz)	1.6	100	2.53	0.799
Mimi	1.1	100	2.28	0.847
EnCodec	1.5	150	1.62	0.651

TABLE II
QUANTIZER ABLATION AT THE MAIN 12.5 Hz / CD64 OPERATING POINT, INTERNAL TRAINING-TIME PROTOCOL. SAME JEPa ENCODER, DECODER ARCHITECTURE, BITRATE, LOSSES AND 400K-STEP SCHEDULE; THE Q2D2 RUN ADDITIONALLY RESUMES FROM A PARTIAL DECODER WARM-START (FIG. 3), SO THIS SHOULD BE READ AS A MATCHED FINAL-OPERATING-POINT COMPARISON RATHER THAN A STRICTLY INITIALIZATION-MATCHED QUANTIZER-ONLY TRIAL.

Quantizer	PESQ	ESTOI
FSQ	2.16	0.825
Q2D2	2.53	0.856

bitrate of 1.1 kbps, attains higher ESTOI (0.847) but lower PESQ (2.275): our codec wins perceptual quality by +0.25 PESQ while trailing on ESTOI by 0.05 at $1.45\times$ the bitrate. DAC and EnCodec-6kbps operate at 4 to $5\times$ our bitrate and are included only as reference upper bounds. We do not claim to beat Mimi; we report the trade-off honestly.

The main model additionally attains UTMOS 3.82 and a word error rate of 23.2% (Whisper base.en, 50 utterances, clean 2026-05-14 run); the higher-bitrate FSQ reference codec reaches 8.6% under the identical protocol. Intelligibility at this bitrate is a limitation we return to in Section V.

Our 237.5 tok/s cd128-FSQ *teacher* (Table I) reaches PESQ 3.14 at ~ 2.9 kbps. The cd64 codec is its 1.6 kbps student; cosine-feature distillation accelerated convergence but did not beat the plain disc-free run.

C. Q2D2 versus FSQ

Table II isolates the quantizer at our main 12.5 Hz, code-dimension-64 operating point. Under the matched final operating point in Table II, replacing FSQ with Q2D2 improves PESQ by +0.37 and ESTOI by +0.031. Q2D2 also converges faster: at 5k steps it reaches PESQ ≈ 1.7 versus FSQ’s ≈ 1.1 . Fig. 3 shows the training curves. The advantage is metric-dependent, however. On the held-out 50-utterance protocol the ranking does not transfer to intelligibility: FSQ reaches a word error rate of 19.4% versus Q2D2’s 23.2%. Q2D2 improves signal-level reconstruction (PESQ, ESTOI) while FSQ better preserves lexical content at this bitrate, a trade-off we report rather than resolve.

Quantizer diagnostics. The 32 per-pair affine transforms remain well-conditioned (mean condition number 1.18, determinants 0.99 ± 0.002), i.e. they adapt without degenerating.

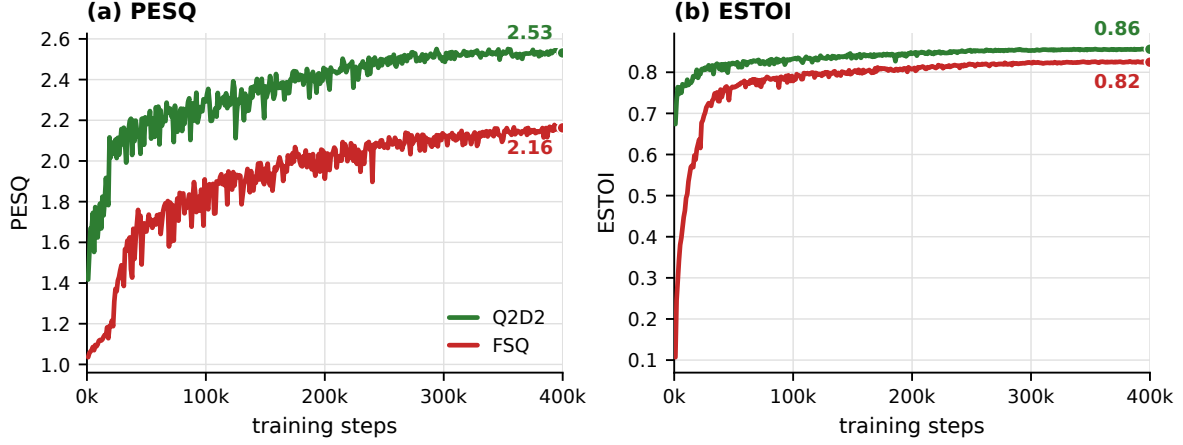


Fig. 3. FSQ vs. Q2D2 at the main 12.5 Hz / cd64 operating point. Both runs use the same Stage-1 JEPA encoder, decoder architecture, bitrate, losses and 400k-step schedule; the Q2D2 run additionally resumes from a partial decoder warm-start, yielding a non-zero starting ESTOI. We therefore interpret the curves as a matched final-operating-point comparison rather than a strictly initialization-matched quantizer-only trial (Table II). Internal training-time eval.

TABLE III

SIGREG ABLATION AT AN AGGRESSIVE 25 Hz / CD32 OPERATING POINT. MATCHED PAIR: IDENTICAL ARCHITECTURE, QUANTIZER AND 100K SINGLE-PHASE TRAINING STEPS; ONLY THE STAGE 1 ENCODER’S SIGREG WEIGHT DIFFERS. INTERNAL TRAINING-TIME PROTOCOL.

Stage-1 encoder	PESQ	ESTOI
with SIGReg ($\lambda_{\text{sigreg}}=0.05$)	2.24	0.793
without SIGReg ($\lambda_{\text{sigreg}}=0$)	1.05	−0.004

The packed token vocabulary of 65,536 is used at 17.0% (entropy ratio 0.77, effective vocabulary $\approx 4,900$), indicating the codec concentrates mass on a meaningful subset rather than collapsing to a few codes or spreading uniformly.

D. SIGReg co-design

Whether the encoder needs SIGReg depends on how hard the quantizer is pushed. Table III reports a matched pair at a harder operating point, 25 Hz and code dimension 32: two Q2D2 codecs identical in architecture, quantizer and 100k single-phase training steps, differing only in whether the Stage 1 encoder was trained with the SIGReg regularizer ($\lambda_{\text{sigreg}}=0.05$) or without it ($\lambda_{\text{sigreg}}=0$). Without SIGReg the codec never leaves the floor (ESTOI -0.004 , PESQ 1.05); with it the same codec trains normally to ESTOI 0.79, PESQ 2.24. Fig. 2 shows the two training curves. The collapse is specific to this aggressive regime: our main 12.5 Hz / cd64 model (Table I) is built on a plain EMA-target encoder with no SIGReg and trains without issue. SIGReg is therefore best read as a co-design tool that becomes necessary as the quantizer is pushed toward lower dimension and higher frame rate, not a universal requirement for pairing Q2D2 with a JEPA encoder.

E. Emergent cross-lingual structure

We extract *pre-quantization* frame-level encoder features for six FLEURS languages (English, French, German, Hindi, Japanese, Chinese), balance to 539 utterances per language,

TABLE IV

LANGUAGE SEPARABILITY OF PRE-QUANTIZATION FRAME-LEVEL FEATURES ON FLEURS (6 LANGUAGES, 5-FOLD CV, CHANCE = 0.167). [E] = ENGLISH-ONLY TRAINING, [M] = MULTILINGUAL. OUR CODEC IS ENGLISH-ONLY AND A 1.6 KBPS RECONSTRUCTION CODEC; SSL MODELS ARE UNCOMPRESSED REPRESENTATION MODELS. UTTERANCE-DISJOINT 5-NN GIVES NEARLY IDENTICAL SCORES: JEPA-EMA 0.850, JEPA-SIGREG 0.791.

Model	Type	LogReg	SVM	5-NN
wav2vec2-base	SSL [E]	0.855	0.847	0.925
Ours (JEPA-EMA)	codec [E]	0.605	0.598	0.854
Ours (JEPA-SIGReg)	codec [E]	0.729	0.715	0.795
XLS-R-300m	SSL [M]	0.685	0.669	0.752
Mimi	codec	0.499	0.489	0.634
mHuBERT-147	SSL [M]	0.484	0.471	0.612
EnCodec	codec	0.405	0.395	0.605
HuBERT-base	SSL [E]	0.409	0.400	0.563
DAC	codec	0.385	0.382	0.451

and measure how separable the languages are under a fair protocol (standardize, PCA to 50 dimensions, equalize frame counts across models). **All FLEURS speech is held out from codec training:** the codec is trained only on LibriLight, and the English FLEURS subset is a corpus disjoint from LibriLight. Every probe, *t*-SNE, NMI score and classifier accuracy in this section is computed on speech the codec has never seen. Table IV reports 5-fold cross-validated classifier accuracy; chance is 0.167. To test whether frame-level 5-NN probes exploit utterance leakage, we repeated the evaluation with utterance-disjoint folds. Scores changed negligibly: JEPA-EMA drops from 0.854 to 0.850, JEPA-SIGReg from 0.793 to 0.791, and the cross-model ordering is unchanged.

Three findings stand out. (i) Among neural codecs ours preserves cross-lingual structure by a wide margin (85.4% vs. Mimi 63.4%, EnCodec 60.5%, DAC 45.1% on 5-NN). (ii) Our English-only 1.6 kbps codec exceeds multilingually-trained SSL (XLS-R 75.2%, mHuBERT 61.2%) and approaches uncompressed English wav2vec2-base (92.5%). (iii)

In this comparison codec architecture appears to dominate training-data multilinguality: multilingually trained EnCodec and DAC carry far less language structure than our English-only codec. We do not claim JEPA uniquely produces cross-lingual structure (wav2vec2-base is better at it, and the phenomenon is documented in prior SSL work); our claim is the *codec* result, with the additional point that our system is the only entry in Table IV that pairs an SSL-style predictive objective with full audio reconstruction.

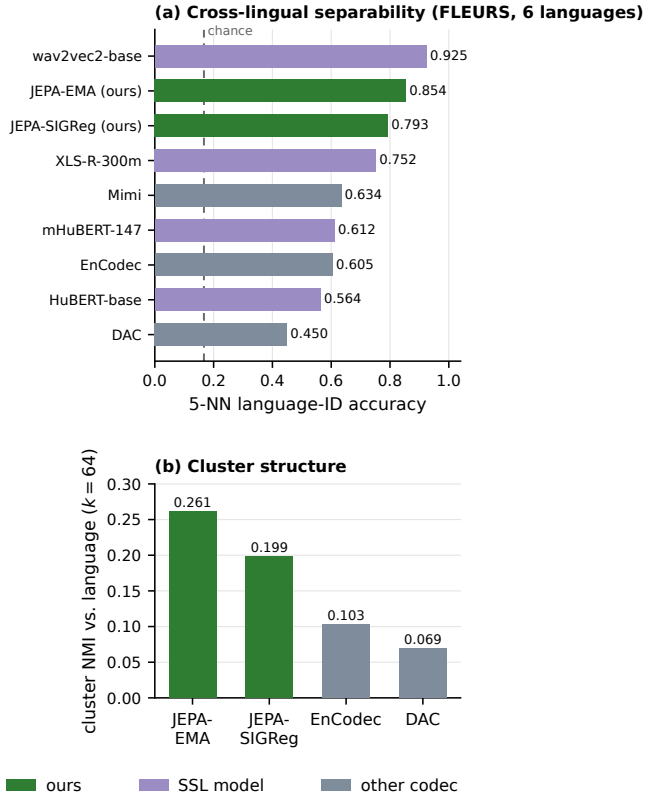


Fig. 4. Cross-lingual structure on six held-out FLEURS languages (five unseen during codec training; English subset disjoint from LibriLight). (a) 5-NN language-ID accuracy: our English-only 1.6 kbps codec exceeds every other neural codec and every multilingually-trained SSL model we test, approaching uncompressed English wav2vec2-base. (b) cluster NMI against language ($k=64$): our encoder forms language-correlated clusters where EnCodec and DAC are diffuse.

Cluster analysis (k -means, $k=64$) supports the same picture: our encoder has the highest NMI against language labels among codecs (0.26 vs. EnCodec 0.10, DAC 0.07) and a positive silhouette, whereas EnCodec/DAC clusters are near-degenerate. We also repeat the analysis on *speech-only frames* (silence removed by an energy threshold), to rule out that the structure is driven by silence regions: the ordering is preserved and our codec retains the highest NMI (0.236 vs. EnCodec 0.091, DAC 0.058 at $k=64$). Fig. 5 shows the corresponding speech-only frame-level t -SNE and the acoustic-category breakdown: the JEPA codecs visibly partition the six languages while DAC remains diffuse, and labelling each cluster by coarse acoustic descriptors (RMS, ZCR, spectral centroid, F0

via pYIN with confidence masking) groups the 64 clusters into five interpretable categories. The encoder’s language structure is thus carried by acoustic-functional groupings rather than an opaque code.

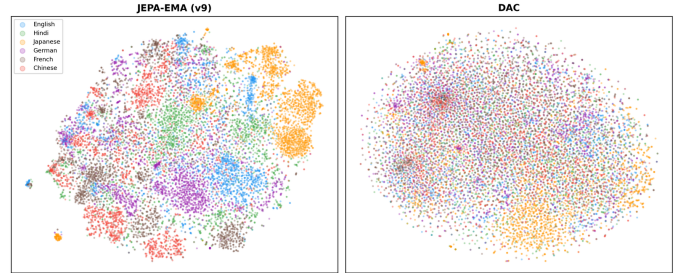


Fig. 5. Speech-only frame-level t -SNE (silence removed), coloured by language. JEPA-EMA (left): the six FLEURS languages form distinct clusters; DAC (right): diffuse, no language structure. The 64 JEPA clusters partition into 15 frication / 15 sonorant-vowel / 14 unvoiced-plosive / 17 mixed / 3 low-energy transition.

F. Style separation across language

As a secondary probe on **held-out Hindi emotional speech** (the codec has seen no Hindi during training), we classify five emotion classes (whisper, angry, excited, neutral, sad; chance 0.20) at utterance level: JEPA-SIGReg reaches 63.4% LogReg, JEPA-EMA 51.3% 5-NN, ahead of EnCodec (37.3%) and DAC (30.8%); on the binary whisper-vs-angry case the gap widens (JEPA-SIGReg 95.8% LogReg, vs. Mimi 80.4%). Fig. 6 shows the utterance-level t -SNE: JEPA codecs cluster emotions where EnCodec and DAC primarily separate only whisper.

IV. RELATED WORK

Neural speech codecs. SoundStream [8] and EnCodec [1] established the RVQ-GAN paradigm; DAC [2] improved it with better quantizer training. Recent low-bitrate codecs include Mimi [3], WavTokenizer [4], BigCodec [9] and SpeechTokenizer [10]. We differ in using a JEPA encoder, a lattice quantizer, and no discriminator.

Quantization. FSQ [5] removes codebook learning with axis-aligned scalar grids; Q2D2 [7] generalizes this to two-dimensional lattices with per-pair affines and was introduced as an audio codec with a conventional encoder. Our contribution is not Q2D2 itself but pairing it with a JEPA encoder and identifying when that pairing needs a Gaussianized source distribution (SIGReg).

Self-supervised speech and JEPA. wav2vec 2.0 [11], HuBERT [12] and WavLM [13] learn representations that, as we and others observe, encode language identity. JEPA [14], [15] predicts masked latents; JEPA-as-tokenizer [6] first used a JEPA encoder for audio coding, which we extend with Q2D2, SIGReg and the cross-lingual analysis.

V. LIMITATIONS

(1) **English-only training.** We use ≈ 5.8 k hours of English-only LibriLight, orders of magnitude less than multilingual

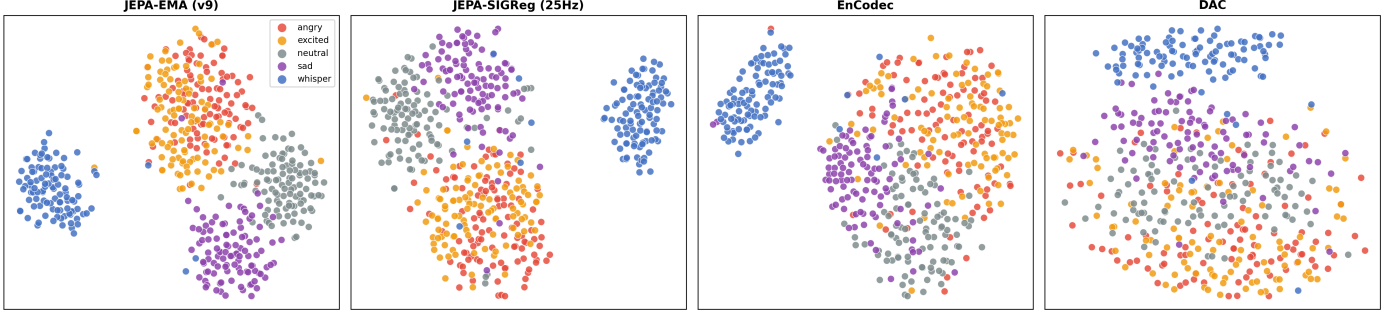


Fig. 6. Five-class Hindi emotion separation, utterance-level t -SNE on a held-out corpus the codec never saw during training. JEPA codecs (ours; left two panels) form per-emotion clusters; EnCodec and DAC primarily separate whisper from the rest. Five classes, balanced; codec trained on English LibriLight only.

SSL (XLS-R 436kh, 128 languages). This is a deliberate sample-efficiency demonstration; multilingual scale-up is future work. (2) **Reconstruction trails Mimi**. On the matched protocol Mimi attains higher ESTOI at a lower bitrate; we win PESQ but the comparison is not bitrate-matched. (3) **Intelligibility**. Word error rate is 23.2% at 1.6 kbps; the higher-bitrate reference codec reaches 8.6% under the identical protocol. Low-frame-rate codecs have a known intelligibility ceiling, and the FSQ variant preserves lexical content better than Q2D2 at this bitrate (19.4%). (4) **wav2vec2-base beats us on cross-lingual separability**; our claim is restricted to “best among codecs” and “competitive with uncompressed SSL while being a 1.6 kbps codec.” (5) **Acoustic categories are heuristic** (signal descriptors, not forced alignment) and no bitrate-matched comparison to BigCodec / WavTokenizer / SemantiCodec is included; only Mimi was run as a low-bitrate baseline.

VI. CONCLUSION

We presented a 1.6 kbps JEPA speech codec built on Q2D2 lattice quantization and trained without adversarial losses. At the main 12.5 Hz / cd64 operating point, Q2D2 improves PESQ over FSQ by +0.37 under a matched internal ablation protocol; separately, a 25 Hz / cd32 experiment shows that SIGReg can prevent Q2D2 collapse when the bottleneck is pushed harder, a concrete instance of distribution-quantizer co-design. On held-out FLEURS speech, the English-trained codec’s pre-quantization encoder features exhibit stronger cross-lingual acoustic structure than every other neural-codec baseline we test, with the structure carried by interpretable acoustic categories. We hope the quantizer recipe and the analysis are useful beyond this specific codec.

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